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Course name	Computer Vision
CO-AT	CO2 - AT1 - Analytical Problem
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1) Thresholding and Intensity level slicing - PCB inspection

Given pixel values:

$r = [25, 60, 95, 130, 165, 200, 235]$

Intensity band = 90-170, display range = 0-255

a) Binary threshold transformation:

$$s = \begin{cases} 255, & r \geq 128 \\ 0, & r < 128 \end{cases}$$

Applying the threshold to each pixel.

input r	output s
25	0
60	0
95	0
130	255
165	255
200	255
235	255

$s = [0, 0, 0, 255, 255, 255, 255]$

b) Intensity level slicing:

$$s = \begin{cases} 255, & 90 \leq r \leq 170 \\ r, & \text{otherwise} \end{cases}$$

input r	output s
25	25
60	60
95	255
130	255
165	255
200	200
235	235

$s = [25, 60, 255, 255, 255, 200, 235]$

c) number & percentage of highlighted pixels:

pixels highlighted are 95, 130, 165.

no. of highlighted pixels = 3

total pixels = 7

$\% = 3/7 \times 100 = 42.86\%$

4) a) Mean intensity of original & sliced output

$$\bar{x} = \frac{25+60+95+130+165+200+235}{7} = \frac{910}{7} = 130$$

sliced o/p mean:

$$\bar{s} = \frac{25+60+255+255+255+200+235}{7} = \frac{1285}{7} = 183.57$$

c) which transformation is more appropriate?

Intensity level slicing is more appropriate for isolating the solder joints because the joints lie in the intensity range 90-170.

It highlights all three pixels in the reqd intensity band. Other pixels retain their original values.

2) Pieswise - Linear contrast stretching - X-Ray image enhancement

$(r_1, s_1) = (70, 20)$

$(r_2, s_2) = (140, 230)$

I/P pixel values; $r = [30, 70, 90, 110, 140, 200]$

control points are: (0,0), (70,20), (140,230) & (255,255)

a) Slope is $m = \frac{s_2 - s_1}{r_2 - r_1}$

$0 \le r \le 70$; $m_1 = \frac{20-0}{70-0} = 0.286$

$s = 2/7\, r$

$70 \le r \le 140$; $m_2 = \frac{230-20}{140-70} = 3$

$$S = 20 + 3(r-70) = 3r - 190$$

$140 \le r \le 255$; $m_3 = \frac{255-230}{255-140} = 0.2174$

$$S = 230 + 5/23 (r-140)$$

b) transformed pixel values:

Given i/p: [30, 70, 90, 110, 140, 200]

i/p r	Calculation	output S
30	(2/7) 30	8.57
70	20	20
90	3(90)-190	80
110	3(110)-190	140
140	230	230
200	230+(5/23)60	243.04

c) Bandwidth & contrast gain

140 - 70 = 70

transformed bandwidth:

230 - 20 = 210

contrast gain = $\frac{210}{70} = 3$

d) compression & expansion:

segment 1: compressed, since slope 0.286 < 1

segment 2: expanded, since slope 3 > 1

segment 3: compressed, since slope 0.2174 < 1

e) Assessment:

The transformation improves visibility of diagnostic details b/w

grey levels to 0 & 140 by increasing their contrast threefold. The trade-off is reduced contrast in the darker and brighter ranges, which may hide some details outside the diagnostic band

3) Spatial filtering - Impulse noise removal

Own image : $\begin{bmatrix} 52 & 55 & 58 \\ 54 & 255 & 57 \\ 53 & 56 & 59 \end{bmatrix}$

a) Mean $= \frac{52+55+58+54+255+57+53+56+59}{9}$

$= \frac{699}{9} = 77.67$

b) weighted - avg filter :

Mask : $\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$ $= (52+110+58+108+1020+114+53+112+59)\frac{1}{16}$

$= \frac{1686}{16} = 105.375$

c) 3 x 3 median filter,

In ascending order : 52, 53, 54, 55, 56, 57, 58, 59, 255

median output = 56

d) absolute error of each filter :

$R = \frac{52+55+58+54+57+53+56+59}{8} = \frac{444}{8} = 55.5$

absolute error : $E = |(\text{filter o/p} - R)|$

$E_{mean} = 77.67 - 55.5$

$= 22.17$

$E_{weighted} = 105.375 - 55.5$

$= 49.875$

E median = 56 - 55.5

= 0.5

2) Most appropriate filter:

Median filter is the most appropriate for removing salt & pepper noise while preserving edges.

It removes the extreme value 255 effectively.

Its output is close to the noise free reference of 55.5

It generally preserves edges better than mean filtering.

4) Edge detection - Sobel and prewitt gradient analysis:

$$I = \begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 90 \\ 10 & 90 & 90 \end{bmatrix}$$

a) Sobel operator:

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \qquad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

$G_x = (90-10) \times 2 + (90-10) + (-10+10)$

$= 160 + 80 = 240$

$G_y = -(10+20+10) + (10+180+90)$

$= -40 + 280 = 240$

b) Gradient magnitude & % error.

$$M = \sqrt{G_x^2 + G_y^2} = \sqrt{240^2 + 240^2}$$

$$= \sqrt{115200} = 339.41$$

$M_{approx} = G_x + G_y = 240 + 240 = 480$

$$E = \frac{M_{approx} - M}{M} \times 100$$

$$E = \frac{480 - 339.41}{339.41} \times 100$$

$$= 41.42\%$$

c) Gradient direction & edge orientation

$$\alpha = \tan^{-1}\left(\frac{G_y}{G_x}\right)$$

$$= \tan^{-1}\left(\frac{240}{240}\right)$$

$$\alpha = 45^\circ$$

d) Horizontal gradient,

$$G_x = 0 + (-10 + 90) + (-10 + 90)$$

$$= 160$$

Vertical gradient,

$$G_y = -(10 + 10 + 10) + (10 + 90 + 90)$$

$$= -30 + 190$$

$$= 160$$

Magnitude, $M = \sqrt{160^2 + 160^2} = 226.27$

$$M_{approx} = 160 + 160 = 320$$

$$E = \frac{320 - 226.27}{226.27} \times 100 = 41.42\%$$

e) Edge detection using threshold : $M \geq 250$

Sobel detects an edge because its magnitude exceeds 250. Prewitt does not detect an edge because its exact magnitude is below 250.

This shows that the approximation can affect threshold-based detection decisions.

5) Global thresholding - Iterative and otsu segmentation

Own histogram:

Intensity n_K	0	1	2	3	4	5	6	7
frequency n_K	3	6	4	1	1	5	8	4

Total pixels = 32

a) global mean intensity,

$$m_G = \frac{\sum n_K n_K}{\sum n_K}$$

$$= \frac{0(3) + 1(6) + 2(4) + 3(1) + 4(1) + 5(5) + 6(8) + 7(4)}{32}$$

$$= \frac{122}{32} = 3.8125$$

b) Iterative global thresholding

$$T_0 = m_G = 3.8125$$

$$m_1 = \frac{0(3) + 1(6) + 2(4) + 3(1)}{3+6+4+1} = \frac{17}{14} = 1.2143$$

$$m_2 = \frac{4(1) + 5(5) + 6(8) + 7(4)}{1+5+8+4} = \frac{105}{18} = 5.833$$

$$T_1 = \frac{m_1 + m_2}{2} = \frac{1.2143 + 5.833}{2} = 3.5238$$

$$T_1 - T_0 = 3.5238 - 3.8125$$

$$= 0.2887 < 0.5$$

$$T = 3.5238$$

c) $\sigma^2_B(k) = P_1(k) P_2(k) [m_1(k) - m_2(k)]$

$K = 2$: $P_1 = \frac{13}{32}$, $P_2 = \frac{19}{32}$; $\sigma^2_B(2) = 5.1202$

$m_1 = \frac{14}{13} = 1.0769$; $m_2 = \frac{108}{19} = 5.6842$

$K = 3$; $P_1 = \frac{14}{32}$, $P_2 = \frac{18}{32}$ $\sigma^2_B(3) = 5.2506$

$m_1 = \frac{17}{14} = 1.2143$; $m_2 = \frac{105}{18} = 5.833$

$K = 4$; $P_1 = \frac{15}{32}$; $P_2 = \frac{17}{32}$ $\sigma^2_B(4) = 5.1354$

$m_1 = \frac{21}{15} = 1.4$; $m_2 = \frac{101}{17} = 5.9412$

d) $N_{obj} = 3 + 6 + 4 + 1 = 14$

$P_{obj} = \frac{14}{32} \times 100$

$= 43.75\%$

e) Compare :

Iterative global thresholding - 3.5238 - 0-3

otsu's method - 3 - 0-3

Both methods separate the image into the same 2 classes. Dark objects and a bright conveyor belt.